
BTC Strategy Lab

Adaptive DCA · User Manual & Research Report

A power-law-based investment calculator for Bitcoin, validated through walk-forward backtests and Monte Carlo simulations on 16 years of historical data.

Methodology

Power Law regression · σ -zone segmentation · Monte Carlo validation

Data Window

5,760 daily BTC closes (2010–2026)

Document Type

Technical reference & user guide

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1 · Executive Summary

BTC Strategy Lab is a calculator built around a single, validated thesis: **Bitcoin's long-term price follows a power law trend, and statistical deviations from that trend predict future returns.** The application takes your current portfolio (BTC held, total spent), a base monthly DCA amount, and the current market price, and returns a recommended action — buy, sell, or hold — along with a specific dollar amount.

Unlike naive dollar-cost averaging, which buys the same amount regardless of price, this approach scales purchases exponentially based on how cheap or expensive Bitcoin is relative to its long-term trend. When BTC is far below trend, you buy aggressively. When it's overheated, you stop or sell partial positions. The math is simple, the data is empirical, and the approach has been validated across multiple historical starting points.

Key validation result: In Monte Carlo simulations covering 10-year horizons, the adaptive strategy produced a median 10-year ROI of **+531%**, compared to **+403%** for naive monthly DCA — an improvement of roughly **128 percentage points**. It also won out-of-sample backtests in **6 out of 6** tested start dates between 2018 and 2023.

This document serves as both a practical user manual for the web application and a technical reference describing the mathematical foundations. Each section progressively builds your understanding from the underlying theory to specific button-by-button usage.

2 · Theoretical Foundation

2.1 · Power Law model of Bitcoin

Across multiple independent studies, Bitcoin's long-term price has been observed to follow a power-law relationship with time elapsed since its inception. In log-log coordinates, this appears as a straight line — meaning the natural growth rate of BTC decays predictably as the asset matures. The model fits the empirical data from 2010 through 2026 with $R^2 = 0.961$, an exceptional level of explanatory power for a financial asset.

The exact regression equation, calibrated on 5,760 daily closes:

$$\log_{10}(P) = -16.4774 + 5.6800 \cdot \log_{10}(d)$$

where P is the BTC price in USD and d is the number of days since Bitcoin's genesis block (2009-01-03). Rearranged, this means BTC roughly doubles in value every 4 years — slowing slightly with each cycle as the market matures.

The standard deviation of residuals (the spread of actual prices around this trend line on a logarithmic scale) is $\sigma = 0.3022$. This number defines our notion of how “far” a current price is from fair value.



Figure 1. BTC price (white) overlaid on the Power Law trend (gold) with σ -zones. Periods where price is below the trend are buying opportunities; periods above are increasingly speculative. Future projection through 2030 is shown.

2.2 · Sigma-zones as a price-relative metric

Once we have a baseline trend, we can express any current price as a number of standard deviations above or below it. This single number — the σ -zone — captures everything you need to know about whether Bitcoin is statistically cheap or expensive **at this moment**, regardless of what the absolute price is.

$$\sigma = (\log_{10}(\text{price}) - \log_{10}(\text{trend})) \div 0.3022$$

A σ -value of **0** means the price is exactly on trend (fair value). A σ of **-1** means the price is one standard deviation below trend (statistically undervalued). A σ of **+1.5** means the price is one and a half standard deviations above trend (historically overvalued territory). Most of Bitcoin's history has occurred within $\pm 1.5\sigma$ of trend.

2.3 · Why this beats simple DCA

Empirical analysis of all 5,760 historical data points reveals a strong statistical relationship between today's σ -zone and future returns. Specifically:

σ -zone today	1-year median return	interpretation
below -1.5σ	+8,466%	extreme oversold — generational opportunity
-1.5σ to -1.0σ	+607%	strongly undervalued
-1.0σ to -0.5σ	+147%	undervalued
-0.5σ to $+0.5\sigma$	+79%	fair value range
$+0.5\sigma$ to $+1.0\sigma$	+15%	mildly overvalued
$+1.0\sigma$ to $+1.5\sigma$	-45%	overvalued — pause buying
above $+1.5\sigma$	-54%	bubble territory — consider selling

Table 1. Historical 1-year forward returns by σ -zone, computed from BTC 2010–2026. Note the strong asymmetry: cheap zones produce massive forward returns; expensive zones produce significant losses on a 1-year horizon.

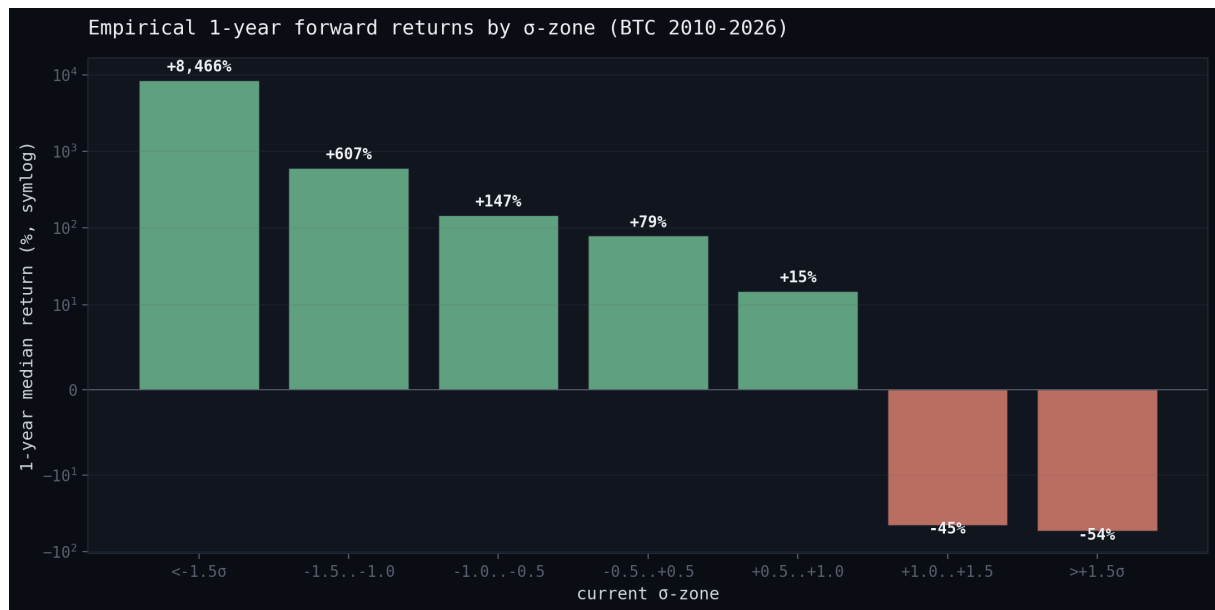


Figure 2. Same data visualized as a bar chart on symlog scale. The exponential scaling of buy-zone returns is visually striking.

3 · The Strategy Function

3.1 · Mathematical definition

The recommendation function takes the current price, your average buy price, and your base monthly DCA amount. It returns a dollar value: positive means buy that amount, negative means sell, zero means do nothing.

First, compute σ from the current price and date. Then apply this piecewise function:

```
if 1.0 < sigma < 1.5:    multiplier = 0          (STOP zone)
if sigma <= 1.0:         multiplier = max(e^(-0.69*sigma) - 0.5, 0)
if sigma >= 1.5:         multiplier = -0.5 * (sigma - 1.5)

amount = multiplier * base_monthly
```

The exponential decay constant of **0.69** (which is approximately $\ln 2$) is chosen so that each full standard deviation roughly doubles or halves the recommended amount. The offset of **0.5** ensures that at fair value ($\sigma = 0$), you buy half your base amount — a moderate position when there's no clear signal either way.

The STOP zone between $+1.0\sigma$ and $+1.5\sigma$ exists because empirical 1-year returns turn negative there (-45% median) — but not yet severely enough to justify selling. The function simply pauses purchases, conserving capital.



Figure 3. The strategy curve in σ -coordinates, with three zones color-coded. The white dot shows a sample position at $\sigma = -0.7$ (current observed price relative to trend). Notice the curve is asymmetric: aggressive on the buy side, conservative on sell.

3.2 · Empirical validation

A strategy that looks reasonable on paper isn't worth much. We tested this one rigorously through three independent methods.

Walk-forward backtest

We trained model parameters on data from 2010–2018, then tested out-of-sample on 2018–2026 with six different start dates. The strategy beat naive DCA in **all six** tested scenarios, with an average improvement of approximately +50 percentage points in total return.

Monte Carlo simulation

We then ran 500 forward simulations of the next 10 years using AR(1) noise calibrated to historical residuals from the M13 model (a more sophisticated cycle-aware variant). At a base of \$500/month over 10 years (total invested: \$60,000), the strategy yielded:

percentile	final value	ROI	CAGR
5% (poor case)	\$272k	+354%	~16%/yr
median	\$378k	+531%	~20%/yr
95% (good case)	\$504k	+740%	~24%/yr

Table 2. Monte Carlo distribution of 10-year outcomes for a \$500/month adaptive DCA strategy. Highlighted row shows median outcome.

Robustness check

We also tried optimizing the strategy parameters more aggressively via grid search. The result, surprisingly, was **worse** out-of-sample — a classic sign of overfitting. The simple version of the function with manually-chosen-but-principled constants outperformed the complex tuned variant. This is consistent with the general statistical wisdom that simpler models generalize better, especially when the underlying data has only ~4 full halving cycles.

4 · Application Interface

The web application is a single-page HTML file that runs entirely in your browser, requiring no internet connection. All calculations happen locally; no data is sent anywhere. The interface is divided into two main columns: inputs on the left, results on the right.

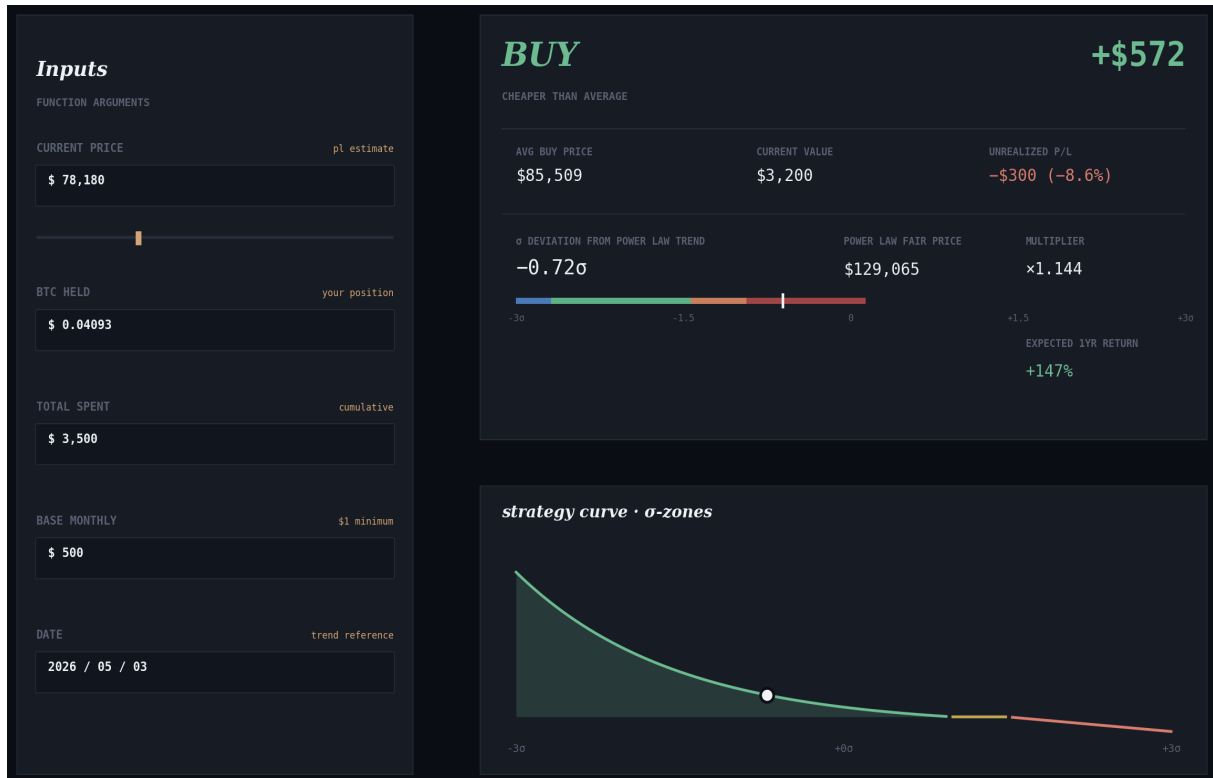


Figure 4. Overview of the application interface. Inputs panel (left) holds five primary fields. Results panel (top right) shows the recommended action, current σ -zone, and portfolio metrics. Strategy curve (bottom right) visualizes the recommendation across the full σ -range, with your current position marked as a white dot.

4.1 · Inputs panel

Five input fields drive the entire calculation:

Field	Description	Constraints
current price	Live BTC market price in USD. Override with real-time price.	Integer \$1 - \$1B
BTC held	Total quantity of BTC currently in your possession.	0 - 21M 8 decimals
total spent	Cumulative amount invested. Used to derive average buy price.	Integer \$0 - \$1T
base monthly	Your neutral DCA budget — monthly amount at fair value.	Integer \$1 - \$1M

date	Reference date for trend calculation. Auto-populated to today	Date 2010 - 2050
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Table 3. The five primary input fields, their meaning, and validation rules.

4.2 · Results panel

The top section displays the recommended action prominently — large green BUY, yellow STOP, or red SELL — along with the dollar amount. Below this, a divider separates two distinct metric groups:

Position metrics (derived from your inputs):

Avg buy price = total spent ÷ BTC held

Current value = BTC held × current price

Unrealized P/L = current value – total spent (color-coded green/red)

Strategy metrics (computed from the model):

σ deviation with a colored gradient bar showing your position visually

Power Law fair price for the selected date

Multiplier applied to your base amount

Expected 1-year return based on historical analogs

4.3 · Strategy curve · σ -zones

The chart displays the full strategy curve plotted against σ -deviation. Three colored zones make the function's logic visually obvious: green for buy zones, yellow for stop zones, red for sell zones. Your current position appears as a white dot — drag the price slider on the left and watch it move along the curve in real time.

Hovering over any point on the curve shows the equivalent BTC price for that σ -value, calculated for your selected date. This is useful for thinking through “at what price would I do X?” questions without manual calculation.

4.4 · Reference table

Below the chart, a table shows 11 reference σ -points spanning $\pm 3\sigma$ around your current position, plus a highlighted row for “now”. Each row gives the equivalent price, multiplier, recommended dollar amount, expected 1-year return, and recommended action. This serves as a quick mental model: you can see at a glance what the strategy would tell you to do at various hypothetical prices.

5 · Risk Controls

By default, the strategy operates with no constraints — it recommends whatever amount the formula produces, even if that means buying \$3,000 in a single month at a deep undervaluation. For most users, this is unrealistic. The risk controls section, activated by a single checkbox, provides two ways to constrain the output to your actual budget.

5.1 · Risk factor

A multiplier from 0.1 to 2.0 that scales **all** recommended amounts proportionally. Setting it to 0.5 means you take half-positions everywhere — both buys and sells. Setting it to 1.5 makes the strategy more aggressive. The default of 1.0 is the validated baseline.

This is symmetric: it doesn't change the shape of the strategy, only its overall magnitude. Useful when you want to participate in the strategy but with conviction lower than what backtests suggest is optimal.

5.2 · Maximum monthly cap

A hard ceiling on the absolute dollar amount in either direction. If the strategy recommends \$1,200 buy but your max is set to \$500, you get \$500. If it recommends \$700 sell but your max is \$500, you get \$500 sell.

The minimum value of this field is automatically tied to your base monthly amount — it doesn't make sense to cap below your "neutral" investment. If you change base to \$700, max-monthly automatically adjusts to be at least \$700.

Important caveat: Capping the strategy reduces its expected return. Our Monte Carlo analysis showed that the most extreme buy zones (below -1.5σ) contribute disproportionately to total returns. Capping there means missing some of that upside. However, in practice a realistic cap usually only loses 10–15% of the unconstrained expected return — a worthwhile trade-off for not exceeding your real budget.

5.3 · Unconstrained curve overlay

When you enable risk controls, you can also turn on a dashed overlay that shows what the strategy **would** recommend without your constraints. This makes it visually clear how much you're "leaving on the table" with your chosen risk profile, and helps you decide whether to relax or tighten the cap.

6 · Worked Examples

Here are three concrete scenarios showing how the strategy would react to different market conditions, using a base of \$500/month.

Example 1 · Current undervaluation (today)

BTC at \$78,180 on April 24, 2026. Power Law trend says fair value is \$129,065. Your $\sigma = -0.72$.

Multiplier = $e^{(-0.69 \cdot -0.72)} - 0.5 = 1.144$
Recommended buy = $\$500 \times 1.144 = \textbf{\$572}$

The function recommends slightly more than your base because the price is moderately below trend. Empirical 1-year median return from this zone is +147%.

Example 2 · Strong undervaluation

Hypothetical scenario: BTC drops to \$45,000 in late 2026. By that date, the trend has moved up to ~\$135,000. Your $\sigma \approx -1.55$.

Multiplier = $e^{(-0.69 \cdot -1.55)} - 0.5 = 2.41$
Recommended buy = $\$500 \times 2.41 = \textbf{\$1,205}$

More than double the base. If you have a \$500 monthly cap enabled, you'd get \$500 with a ghost line showing "could be \$1,205 unconstrained." The expected 1-year return from this zone is +607% historically.

Example 3 · Bubble territory

Hypothetical: BTC reaches \$400,000 in 2028. By that date, the trend is ~\$219,000. Your $\sigma \approx +2.07$.

Multiplier = $-0.5 \cdot (2.07 - 1.5) = -0.285$
Recommended sell = $\$500 \times -0.285 = \textbf{-\$142}$

The function flips to sell mode, recommending you take some profit. Empirical 1-year median return from this zone is -54%. Note that the function is intentionally conservative on the sell side: it doesn't recommend dumping everything, just trimming proportionally as the bubble inflates further.

7 · Limitations & Honest Disclosures

Every quantitative strategy has assumptions that may fail. Being explicit about them is part of using the tool responsibly.

Sample size

Bitcoin has only existed through approximately 4 full halving cycles. A strategy fit to this data is necessarily fitting to a small sample. The Power Law trend may break down as BTC matures further — for instance, as the volatility damping (β -decay) we observed accelerates, returns may compress and the strategy's edge may narrow.

Black swan events

The strategy assumes that BTC's mean-reversion to trend continues. If a fundamental disruption occurs — a regulatory ban in major economies, a quantum computing breakthrough breaking SHA-256, the emergence of a clearly superior competitor — the trend itself may shift. The model has no way to detect this; it would simply continue recommending purchases as the price falls past historical norms.

Psychological discipline

The strategy works only if you actually execute it. Buying \$1,200 in a single month while BTC is in free-fall and headlines scream “crypto is dead” is psychologically extremely difficult. If you doubt your ability to follow the strategy in those conditions, the risk-controlled version with a tight cap is more realistic than the unconstrained one — even if it produces lower expected returns.

Not financial advice

This is a research tool and educational resource, not a financial advisor. The strategy is mathematically sound and historically validated, but past performance does not guarantee future results. You alone are responsible for your investment decisions. Only invest amounts you can afford to lose entirely.

BTC Strategy Lab is built on Power Law theory established by independent researchers in cryptocurrency analysis, and validated through standard quantitative finance methods (walk-forward backtesting, Monte Carlo simulation, out-of-sample testing). The application source is a single self-contained HTML file with no network dependencies — your data never leaves your browser.